Writing maths on screen

A guide for students answering papers on Prep Portal. You type the maths the way you would say it, and the box sets it properly, the way it looks in your textbook.

1 Opening the maths box

1. Click in the answer where the maths should go.
2. Press `Alt` + `=`. A small box opens with a preview under it.
3. Type the maths, the way it is said. The preview shows what you will get.
4. Press `Enter` to put it in your answer. `Esc` closes the box without it.
5. To change it later, double-click the formula. It opens again with what you typed.

Words outside the box are written in handwriting, the way your answers look on paper. Keep the box for maths.

2 Fractions

YOU TYPE	YOU GET	SAY IT
`1/2`	$\frac{1}{2}$	one over two
`2x/3`	$\frac{2x}{3}$	two x over three
`(x + 1)/2`	$\frac{x+1}{2}$	x plus one, all over two
`a/(b + c)`	$\frac{a}{b+c}$	a over b plus c
`1/2 + 1/3`	$\frac{1}{2} + \frac{1}{3}$	a half plus a third
`x + 1/2`	$x + \frac{1}{2}$	x plus a half

THE TRAP The slash takes only what came since the last plus or minus. For x plus one ALL over two, put brackets round the top: `(x + 1)/2`. The brackets disappear once it is set.

3 Mixed numbers

YOU TYPE	YOU GET	SAY IT
`2 1/3`	$2\frac{1}{3}$	two and a third
`3 3/4 - 1 1/4`	$3\frac{3}{4} - 1\frac{1}{4}$	three and three quarters take away one and a quarter
`3 3/4 div 1 1/2`	$3\frac{3}{4} \div 1\frac{1}{2}$	three and three quarters divided by one and a half

THE TRAP The SPACE between the whole number and the fraction matters. `21/3` is twenty-one thirds, not two and a third.

4 Powers

YOU TYPE	YOU GET	SAY IT
`x^2`	x^2	x squared
`x^2 - 5x + 6 = 0`	$x^2 - 5x + 6 = 0$	a quadratic equation
`2^(n + 1)`	2^{n+1}	two to the power n plus one
`10^(-3)`	10^{-3}	ten to the power minus three
`x^(1/2)`	$x^{\frac{1}{2}}$	x to the power a half
`(x + 1)^2`	$(x+1)^2$	x plus one, all squared
`1.5 xx 10^3`	1.5×10^3	standard form

THE TRAP The hat `^` lifts only the ONE thing after it. `2^n + 1` gives $2^n + 1$. A power longer than one character goes in brackets.

5 Roots

YOU TYPE	YOU GET	SAY IT
`sqrt(9)`	$\sqrt{9}$	the square root of nine
`sqrt(b^2 - 4ac)`	$\sqrt{b^2 - 4ac}$	the root of b squared minus four a c
`cbrt(27)`	$\sqrt[3]{27}$	the cube root of twenty-seven
`root(4)(16)`	$\sqrt[4]{16}$	the fourth root of sixteen

6 Small low numbers and bases

YOU TYPE	YOU GET	SAY IT
`x_1`	x_1	x one
`a_(n + 1)`	a_{n+1}	a, n plus one
`1011_2 = 11_10`	$1011_2 = 11_{10}$	one zero one one base two is eleven base ten
`45_b`	45_b	forty-five base b

REMEMBER The underscore `_` drops the next thing LOW. Write the base on every line where it matters, or the marker cannot tell the bases apart.

7 Logarithms

YOU TYPE	YOU GET	SAY IT
`log_2(8) = 3`	$\log_2(8) = 3$	log to the base two of eight is three
`log(100)`	$\log(100)$	log of a hundred
`ln(x)`	$\ln(x)$	natural log of x

8 Signs

YOU TYPE	YOU GET	YOU TYPE	YOU GET
`3*4`	3×4	`12 div 4`	$12 \div 4$
`3 xx 4`	3×4	`2 cdot 5`	$2 \cdot 5$
`x <= 5`	$x \leq 5$	`y >= 2`	$y \geq 2$
`a != b`	$a \neq b$	`pi ~= 3.14`	$\pi \approx 3.14$
`x = +- 3`	$x = \pm 3$	`x -> 0`	$x \to 0$
`45%`	45%	`therefore x = 2`	$\therefore x = 2$

REMEMBER The slash makes a fraction. For the division sign itself, type the word `div`.

9 Sets

YOU TYPE	YOU GET	YOU TYPE	YOU GET
`x in A`	$x \in A$	`emptyset`	$\varnothing$
`A cup B`	$A \cup B$	`A cap B`	$A \cap B$

10 Greek letters and functions

YOU TYPE	YOU GET	YOU TYPE	YOU GET
`pi`	π	`theta`	θ
`alpha + beta`	$\alpha + \beta$	`Delta x`	Δx
`f(x) = 2x + 1`	$f(x) = 2x + 1$	`abs(x - 3)`	$\lvert x - 3 \rvert$
`sin(x)`	$\sin(x)$	`tan(theta)`	$\tan(\theta)$
`cos(30 deg)`	$\cos(30^\circ)$	`cos(x)`	$\cos(x)$

11 Geometry

YOU TYPE	YOU GET	YOU TYPE	YOU GET
`90 deg`	90°	`angle ABC`	$\angle ABC$
`triangle ABC`	$\triangle ABC$	`vec(AB)`	$\vec{AB}$

12 Bigger pieces

YOU TYPE	YOU GET	SAY IT
`sum_(i=1)^n i`	$\sum_{i=1}^{n} i$	the sum of i, from i equals one to n
`int_0^1 x dx`	$\int_0^1 x\,dx$	the integral of x from nought to one
`sin^2(x) + cos^2(x) = 1`	$\sin^2(x) + \cos^2(x) = 1$	sine squared x plus cosine squared x is one
`binom(5)(2)`	$\binom{5}{2}$	five choose two
`bar(x)`	$\overline{x}$	x bar, the mean
`pmatrix(1, 2; 3, 4)`	$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$	a matrix: commas between, a semicolon for each new row

BEFORE YOU PRESS ENTER Look at the preview. If something is in the wrong place, press Backspace and add brackets. The preview is there so you catch it first.